Horses and Math Lesson Plan

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Two hours, four kids aged 6 to 12, one calm horse, one mechanical metronome, and one question for the day: **Can we find the horse's beat?** First with its feet, then with its heart. A break comes right before each piece of equipment goes on.

Setup before the kids arrive

- [ ] Mechanical metronome, wound up; check the weight slides to 60, 100 and 144
- [ ] Stopwatch
- [ ] 4 hobby horses
- [ ] Chalk or cones for a 20 foot runway the horse walks past
- [ ] Print page 2 (Heartbeat Chart) once, big, on a clipboard, with markers
- [ ] Polar Equine strap, water to wet it, and the phone app signed in; it goes on during the second break
- [ ] Snack and water for the two breaks
- [ ] Stethoscope as a backup if the Polar will not read

Give the 12-year-old a job before anything starts: lab assistant, who runs the metronome and writes the numbers on the big chart. That keeps them engaged without the lesson feeling babyish.

The two hours

Time	Block	What happens	Math hook
0:00 to 0:15	Four feet, two feet (at the fence)	Ask the day's question: can we find the horse's beat? Kids count their own footsteps across the room: 2 sounds per stride. Watch the horse walk and clap each hoof hit: 4 beats.	Counting, 2 versus 4

Time	Block	What happens	Math hook
0:15 to 0:30	Hobby-horse gaits	Walk = 4 flat steps, say "one-two-three-four." Trot = bounce "one-two, one-two." Canter = "da-da-DUM" skip. These are approximations, so ask how their gait is different: they have 2 legs, so their walk has 2 beats and the horse's has 4. Which of their own gaits has 2 beats? Can they make 3? You call a gait, they switch; then the assistant calls.	Beat patterns 4, 2, 3
0:30 to 0:45	Metronome stepping	Assistant slides the weight to 60, then 100, then 144. Everyone steps to it. Which feels like walk, trot, canter? The group picks a number for each gait; assistant writes it on the chart.	Beats per minute; count 10 seconds and times 6; find the halfway number
0:45 to 0:55	Break	Snack, pet a horse, no instructions.	none
0:55 to 1:15	Measure the real horse (by ear)	Adult walks the horse up and down the runway while the kids clap each hoofbeat. The older two count strides for 30 seconds on the stopwatch; times 2 is strides per minute, write it on the chart. Assistant slides the weight until the clicks land on the hoofbeats. Repeat at trot. Everyone steps with the horse.	Count 30 seconds, times 2; ratio: how many kid steps fit in one horse stride?
1:15 to 1:25	Break	Water, grooming brushes out. Adults wet the Polar strap and put it on.	

Time	Block	What happens	Math hook
1:25 to 1:45	Heartbeats (Polar Equine)	Kids find their pulse on neck or wrist, count 10 seconds, times 6, write it on the chart. Show the horse's number on the Polar app. Ask why the big animal is slower. Fill in cow, goat, chicken together. Set the metronome to 40, its slowest, and point out the resting horse is slower than the slowest setting.	Order slowest to fastest; which animal is about 10 times the horse?
1:45 to 1:55	Jump test	20 jumping jacks, retake the pulse, write the new number next to the old one. Show the horse's Polar number after the trot versus at rest.	Subtraction: how much did it go up?
1:55 to 2:00	Wrap	Each kid shows one gait on the hobby horse while the others call the beat count. Answer the day's question together: yes, and here is its number.	

Why the big animal is slower, in kid words: a bigger heart pushes more blood with each squeeze, so it needs fewer squeezes a minute.

If attention runs out

- Any block can be cut in half. Metronome stepping and Heartbeats carry the math, so protect those two.
- Keep a "free gait" minute in your pocket: shout "canter!" and let them run it off, then come back.
- If the Polar will not read, use a stethoscope behind the horse's left elbow. Kids count aloud for 15 seconds, times 4.
- If the horse will not trot in hand, match the metronome to the walk only and do trot and canter on the hobby horses.

- The 6 and 7-year-olds can skip the multiplication. They count beats and tap the rhythm; the older two do the times 6 and times 2.

Heartbeat Chart (page 2, one for the group)

Today's question: Can we find the horse's beat?

My metronome numbers

Gait	Beats in one stride	Metronome setting that feels right	The real horse (strides per minute)
Walk			
Trot			
Canter			

Heartbeats per minute

Who	Count for 10 seconds	Times 6 = beats per minute	After 20 jumping jacks
_____			
_____			
_____			
_____			
The horse (name: _____)			
Cow			
Goat			
Chicken			

Put them in order, slowest heart to fastest

1. _____
2. _____

3. _____
4. _____
5. _____

Which animal's heart beats about 10 times as fast as the horse's? _____

Instructor reference numbers

Measure on the day; these are sanity checks and fill-ins if the Polar fails. A mechanical metronome runs 40 to 208 in fixed steps, so match the horse to the nearest marking.

Gaits

Gait	Beats per stride	Rhythm to say	Horse strides per minute (approximate)	Metronome marking for kids to step to
Walk	4	one-two-three-four	50 to 60 (one of Mermaid's slow walks measured near 35)	60
Trot	2	one-two, one-two	80 to 95	100
Canter	3 plus a pause	da-da-DUM	100 to 110	144

Resting heart rates, beats per minute (approximate)

Who	Resting	Note
Blue whale	about 10	for the gasp
Horse	28 to 44	your Polar sessions: walk 76 to 115, trot 85 to 145, canter 115 to 157
Cow	50 to 80	
Adult person	60 to 100	
Kid aged 6 to 12	70 to 110	
Goat	70 to 90	

Who	Resting	Note
Chicken	250 to 350	about 10 times the horse
Hummingbird	about 1,200	for the second gasp

Polar's smoothed number runs low during exercise, by roughly 8 bpm at trot and 16 at canter, so the horse's after-trot reading on the app will understate the real jump. That is fine for the kids; it still rises clearly.